

FLYBY: MATHEMATICAL FOUNDATIONS OF ANTHROPOMETRIC 3D ELEVATION & FLIGHT TRAJECTORY ACOUSTICS

Physics-Based Pinna Dispersion, Dynamic Transient Preservation, and Boundary Grounding

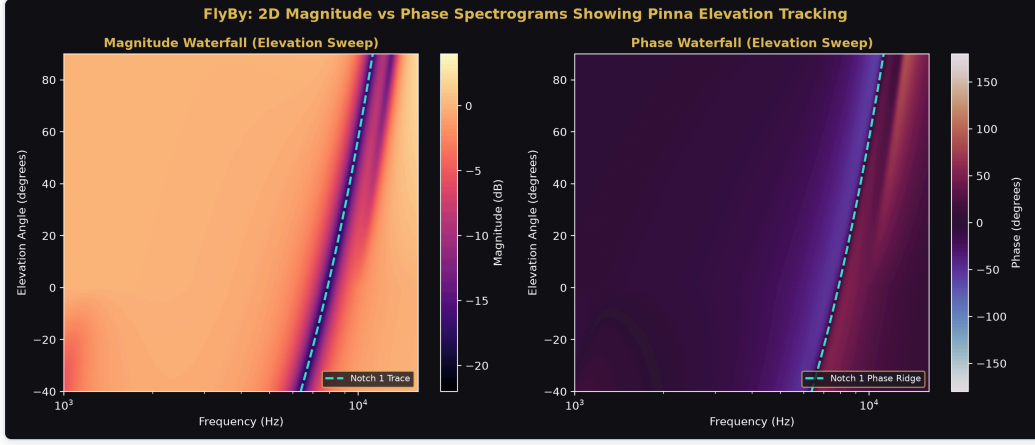
Kijjaz • Gemini 3.8 Flash

Audio DSP

$$\tau_{\text{ITD}}(\theta) = \frac{a}{c}(\sin |\theta_{\text{rad}}| + |\theta_{\text{rad}}|) \cdot \text{sgn}(\theta_{\text{rad}})$$

Delay line reading is implemented with 4-point Cubic Hermite interpolation to guarantee sub-sample smoothness and eliminate zipper noise during high-speed trajectory flybys.

2.2 Rayleigh Head Diffraction Filter



Configuration	Azimuth θ	Elevation ϕ	ITD (μ s)	ILD (dB)	Notch 1 (Hz)	Torso Dip
Pure Horizontal (Full Left)	-90°	0°	+510.2	-9.3 dB	7,857 Hz	0.0 dB
Front Center (Eye Level)	0°	0°	0.0	0.0 dB	7,857 Hz	0.0 dB
Pure Horizontal (Full Right)	$+90^\circ$	0°				